

Apollo Diagnostics - Laboratory Investigation Report

123, Nungambakkam High Road, Chennai - 600034 | NABL Accredited | Tel: 044-2827-5000

Patient Name:	Hemalatha R	Report ID:	APL-AUTO-42-HEM-2026
Age / Gender:	42 Yrs / Female	Category:	Autoimmune / Inflammatory Markers Elevated
Weight / Height:	63 kg / 160 cm	BMI:	24.6 kg/m2
Ref. Doctor:	Dr. Sneha Patel	Report Date:	01-Mar-2026

1. COMPLETE BLOOD COUNT [Circulatory System]

Red Cell Indices

Test	Result	Unit	Reference Range	Flag
Haemoglobin (Hb)	10.4	g/dL	12.0 - 16.0	LOW
Packed Cell Volume (PCV)	39.0	%	36 - 46	Normal
MCV	90.0	fL	83 - 101	Normal
MCH	29.0	pg	27 - 32	Normal
RDW-CV	13.0	%	11.6 - 14.0	Normal

Haematology Comment: Haemoglobin 10.4 g/dL - normocytic anaemia. MCV and MCH within range. Consider anaemia of chronic disease, early iron deficiency or haemolysis.

White Cells & Platelets

Test	Result	Unit	Reference Range	Flag
Total WBC	5347	cells/cmm	4500 - 11000	Normal
Neutrophils	62	%	40 - 75	Normal
Lymphocytes	28	%	20 - 45	Normal
Platelet Count	2.8	lakh/cmm	1.5 - 4.0	Normal

Comment: WBC and platelet counts within reference limits. No significant leucocytosis, leucopenia or thrombocytopenia.

2. IRON STUDIES [Haematopoietic System]

Test	Result	Unit	Reference Range	Flag
Serum Iron	110	ug/dL	60 - 170	Normal
TIBC	310	ug/dL	250 - 370	Normal
Transferrin Saturation	35.0	%	20 - 50	Normal
Serum Ferritin	439.3	ng/mL	12 - 150	HIGH

Iron Studies Comment: Iron stores and transport markers are within normal limits. No iron deficiency or iron overload.

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3. HEPATIC FUNCTION TEST [Hepatobiliary System]

Liver Enzymes

Test	Result	Unit	Reference Range	Flag
SGPT / ALT	22	U/L	7 - 56	Normal
SGOT / AST	20	U/L	10 - 40	Normal
Alkaline Phosphatase (ALP)	70	U/L	44 - 147	Normal
Total Bilirubin	0.6	mg/dL	0.2 - 1.2	Normal
Albumin	3.1	g/dL	3.5 - 5.0	LOW

Hepatologist Note: Liver enzymes within normal limits. No hepatocellular damage or cholestatic process detected.

4. RENAL FUNCTION TEST [Urinary System]

Test	Result	Unit	Reference Range	Flag
Serum Creatinine	1.3	mg/dL	0.6 - 1.1	HIGH
Blood Urea Nitrogen	14	mg/dL	7 - 20	Normal
eGFR (CKD-EPI)	98	mL/min/1.73m2	60 - 120	Normal
Serum Uric Acid	4.8	mg/dL	2.4 - 6.0	Normal
Sodium (Na+)	139	mEq/L	135 - 145	Normal
Potassium (K+)	4.0	mEq/L	3.5 - 5.1	Normal

Nephrologist Note: eGFR 98 mL/min/1.73m2 (reduced) and creatinine 1.3 mg/dL (elevated). Consistent with early to moderate chronic kidney disease. Monitor BP, restrict protein, avoid nephrotoxic drugs. Nephrology referral advised if eGFR <45.

5. LIPID PROFILE [Cardiovascular Risk]

Test	Result	Unit	Reference Range	Flag
Total Cholesterol	172	mg/dL	0 - 200	Normal
HDL Cholesterol	55	mg/dL	50 - 999	Normal
LDL Cholesterol	98	mg/dL	0 - 100	Normal
Triglycerides	88	mg/dL	0 - 150	Normal

Cardiologist Note: Lipid profile within optimal range. Cardiovascular risk from dyslipidaemia is low.

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6. THYROID FUNCTION TEST [Endocrine System]

Test	Result	Unit	Reference Range	Flag
TSH	2.1	uIU/mL	0.4 - 4.0	Normal
Free T3	3.2	pg/mL	2.3 - 4.2	Normal
Free T4	1.1	ng/dL	0.8 - 1.8	Normal

Endocrinologist Comment: Thyroid function normal. TSH 2.1 uIU/mL. Euthyroid state confirmed.

7. DIABETES SCREENING [Endocrine - Pancreas]

Test	Result	Unit	Reference Range	Flag
Fasting Blood Glucose	88	mg/dL	70 - 100	Normal
Post-Prandial Glucose (2hr)	115	mg/dL	0 - 140	Normal
HbA1c	5.1	%	0 - 5.7	Normal
HOMA-IR	1.5		0 - 2.5	Normal

Endocrinology Note: Glucose metabolism normal. HbA1c 5.1% - no diabetes or pre-diabetes. HOMA-IR 1.5 - no insulin resistance.

8. VITAMINS & BONE [Nutritional / Skeletal]

Test	Result	Unit	Reference Range	Flag
Vitamin D (25-OH)	19.1	ng/mL	30 - 100	LOW
Vitamin B12	450	pg/mL	211 - 911	Normal
Folate (Serum)	12.0	ng/mL	5.4 - 999	Normal
Calcium (Total)	9.4	mg/dL	8.5 - 10.5	Normal
PTH	32	pg/mL	15 - 65	Normal

Nutritionist/Endocrinology Note: Vitamin D: 19 ng/mL - deficient (10-20). PTH at 32 pg/mL is within range. Supplementation: 60,000 IU cholecalciferol weekly x 8 weeks, then 1000-2000 IU/day. Recheck in 3 months.

9. INFLAMMATORY MARKERS [Immune System]

Test	Result	Unit	Reference Range	Flag
ESR	80	mm/hr	0 - 20	HIGH
CRP	38.0	mg/L	0 - 5	HIGH
hs-CRP	9.3	mg/L	0 - 1.0	HIGH

Immunology Comment: CRP 38.0 mg/L and ESR 80 mm/hr are elevated. hs-CRP 9.3 mg/L - high CV risk (>3). Inflammatory markers elevated - correlate with clinical symptoms and autoimmune screen.

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CLINICAL SUMMARY & RECOMMENDATIONS

- [ABNORMAL]** Anaemia: Hb=10.4 g/dL (LOW). Normocytic - check iron stores and chronic disease.
- [DEFICIENT]** Vitamin D=19 ng/mL (LOW). Supplementation required.
- [HIGH]** CRP=38.0 mg/L, hs-CRP=9.3 mg/L - systemic inflammation present.
- [ACTION]** Next appointment: Dr. Sneha Patel in 4-8 weeks for follow-up and review of supplementation response.

Report reviewed by: Dr. Anitha Krishnamurthy, MD (Pathology) | Verified by Dr. Sneha Patel

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